Does retrieval practice improve long-term second language vocabulary retention compared to recognition-based tasks?

Yes, **retrieval practice improves** long-term foreign language vocabulary retention compared to recognition-based tasks.

Consensus Meter N = 13



Yes 85% · Possibly 0% · Mixed 15% · No 0%

FIGURE 1 Scientific consensus on retrieval practice enhancing second language vocabulary retention

Empirical evidence consistently shows that actively recalling foreign language vocabulary yields stronger long-term retention than recognition tasks, passive rereading, or repeated studying (Long et al., 2026; Makarchuk, 2018; Nakata, 2016). While recognition tasks provide initial contextual familiarity, recall-based retrieval creates more durable memory traces over extended intervals (Long et al., 2026; Makarchuk, 2018; Van Den Broek et al., 2018).

Comparative Retention Across Task Formats

Direct comparisons between learning formats reveal an advantage for active recall over recognition-based and passive methods (Makarchuk, 2018; Candry et al., 2020). A semester-long investigation comparing rereading, recognition, and recall found that recalling words was significantly more effective for long-term retention, regardless of the subsequent test format (Makarchuk, 2018). Recognition-oriented practice was less effective at consolidating lexical knowledge, particularly when learners were required to produce orthographic forms (Makarchuk, 2018; Nakata, 2016).

Similarly, investigations evaluating word-writing, passive study, and retrieval practice demonstrated that retrieving foreign words based on translation prompts led to superior immediate and delayed recall across one-week intervals (Candry et al., 2020). When evaluated over longer horizons, retrieval practice maintains an advantage over study-only conditions at delays of 30 and 50 days (Long et al., 2026). However, recognition formats remain functional when learners only need receptive lexical access rather than full productive spelling (Nakata, 2016).

Study Design & Target Language	Practice Conditions Compared	Retention Interval	Key Finding
Within-subjects L2 Swahili (Nakata, 2016)	Recall vs. recognition vs. hybrid (Nakata, 2016)	Delayed testing (Nakata, 2016)	Recall optimizes **productive orthography**; recognition aids receptive matching (Nakata, 2016)

Study Design & Target Language	Practice Conditions Compared	Retention Interval	Key Finding
15-week EFL semester cohort (Makarchuk, 2018)	Recalling vs. recognizing vs. rereading (Makarchuk, 2018)	Cumulative semester (Makarchuk, 2018)	Active recall outperforms **recognition learning** across test types (Makarchuk, 2018)
Secondary EFL learners (Candry et al., 2020)	Retrieval vs. word writing vs. study (Candry et al., 2020)	1 week (Candry et al., 2020)	Retrieval practice maximizes **form and meaning** retention (Candry et al., 2020)
Experimental L3 Japanese (Long et al., 2026)	Covert retrieval vs. restudy (Long et al., 2026)	30 and 50 days (Long et al., 2026)	Retrieval practice yields **durable recognition** after 50 days (Long et al., 2026)

FIGURE 2 Summary of empirical studies comparing retrieval practice with alternate vocabulary learning conditions

Underlying Cognitive and Neural Mechanisms

The superior mnemonic impact of retrieval practice is supported by cognitive effort and semantic elaboration accounts (Long et al., 2026; Huang et al., 2025; Van Den Broek et al., 2022). Electrophysiological recordings during post-retrieval recognition tests demonstrate attenuated negative N400 waveforms compared to restudy conditions, reflecting deeper semantic processing during initial retrieval (Li et al., 2023). The cognitive struggle involved in generating a target word reconstructs internal semantic networks, whereas recognition tasks rely heavily on passive perceptual familiarity (Long et al., 2026; Huang et al., 2025; Huang, 2024).

```
flowchart LR
    A[Vocabulary Practice] --> B[Recognition Task]
    A --> C[Retrieval Practice]
    B --> D[Perceptual Familiarity]
    C --> E[Effortful Search & Reconstruction]
    E --> F[Semantic Network Integration]
    D --> G[Rapid Memory Decay]
    F --> H[Durable Long-Term Retention]
```

Both overt spoken retrieval and covert mental retrieval produce equivalent mnemonic gains in vocabulary acquisition (Li et al., 2024). However, providing immediate corrective feedback is crucial; without feedback, uncorrected errors can impair performance relative to repeated exposure (Long et al., 2026; Li et al., 2022).

Key Boundary Conditions and Modulators

The efficacy of retrieval practice relative to other tasks is modulated by several structural variables:

- **Retrieval dosage**: Practicing retrieval only once or twice often fails to outperform restudying, whereas four or more repetitions generate substantial retention gains (Huang et al., 2025; Qu et al., 2024; Miyatsu & McDaniel, 2019; Qu et al., 2024).
- **Mnemonic scaffolding**: Coupling limited retrieval attempts with keyword mnemonics overcomes low initial recall rates and stabilizes long-term retention (Qu et al., 2024; Miyatsu & McDaniel, 2019).
- **Temporal spacing**: Spacing retrieval trials evenly across days introduces desirable difficulty that improves delayed retention compared to massed or early cramming regimens (Karatas et al., 2021; Tuychiev, 2026; Karpicke & Roediger, 2007).
- **Practice variability**: Varying the direction of retrieval prompts produces more flexible vocabulary recall over a 24-hour delay than static retrieval conditions (Soderstrom, 2026).

Table 2. Final test score for all four treatments. (Qu et al., 2024)

In summary, retrieval practice provides a distinct advantage over recognition-based learning for long-term second language vocabulary retention, especially for active recall and orthographic production (Makarchuk, 2018; Nakata, 2016). Maximizing these benefits requires sufficient retrieval frequency, corrective feedback, and spaced practice schedules (Long et al., 2026; Huang et al., 2025; Karpicke & Roediger, 2007).

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at https://consensus.app. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

References

Candry, S., Decloedt, J., & Eyckmans, J. (2020). Comparing the merits of word writing and retrieval practice for L2 vocabulary learning. https://doi.org/10.1016/j.system.2020.102206

Huang, S., Cao, H., Wang, J., & Liu, S.-T. (2025). The Moderating Role of Learning Rounds: Effects on Retrieval Practice and Context Dependence in Digital Flashcard Foreign Language Vocabulary Learning. *Behavioral Sciences, 15*. https://doi.org/10.3390/bs15111540

Huang, Q. (2024). Research on the Influence of Retrieval Practice on ESL Vocabulary Memory. *Communications in Humanities Research*. https://doi.org/10.54254/2753-7064/36/20240044

Karatas, N., Lovelett, J., Demir, B., Erkol, K., Veríssimo, J., Erçetin, G., & Ullman, M. (2021). Improving second language vocabulary learning and retention by leveraging memory enhancement techniques: A multidomain pedagogical approach. *Language Teaching Research, 29*, 112 - 149. https://doi.org/10.1177/13621688211053525

Karpicke, J. D., & Roediger, I. H. (2007). Expanding retrieval practice promotes short-term retention, but equally spaced retrieval enhances long-term retention.. *Journal of experimental psychology. Learning, memory, and cognition, 33 4*, 704-19. https://doi.org/10.1037/0278-7393.33.4.704

Li, J., Zhang, E.-H., Zhang, H.-H., Li, D.-F., & Cao, H. (2023). Neural correlates of retrieval practice on the learning and memory retention of L3 French vocabulary. *International Journal of Multilingualism, 21*, 1369 - 1385. https://doi.org/10.1080/14790718.2023.2166943

Li, J., Cao, H., & Li, D.-F. (2024). Covert and overt retrieval are equally beneficial for learning a third language vocabulary. *International Journal of Multilingualism, 22*, 2553 - 2571. https://doi.org/10.1080/14790718.2024.2443593

Li, J., Zhang, E.-H., Zhang, H.-H., Gou, H., & Cao, H. (2022). Retrieval practice plus feedback benefits a third language vocabulary learning. *International Journal of Multilingualism, 21*, 1014 - 1033. https://doi.org/10.1080/14790718.2022.2102172

Long, S., Wang, Y.-J., Chen, J., Cao, H., & Tang, Y. (2026). The Benefits of Covert Retrieval Practice on Short- and Long-Term L3 Vocabulary Retention. *Behavioral Sciences, 16*. https://doi.org/10.3390/bs16071124

Makarchuk, D. (2018). Recall Efficacy in EFL Learning. *English teaching, 73*, 115-138. https://doi.org/10.15858/engtea.73.2.201806.115

Miyatsu, T., & McDaniel, M. (2019). Adding the keyword mnemonic to retrieval practice: A potent combination for foreign language vocabulary learning?. *Memory & Cognition, 47*, 1328 - 1343. https://doi.org/10.3758/s13421-019-00936-2

Nakata, T. (2016). Effects of retrieval formats on second language vocabulary learning. *International Review of Applied Linguistics in Language Teaching, 54*, 257 - 289. https://doi.org/10.1515/iral-2015-0022

Qu, K., Liu, T., Qiao, Y., & Wang, P. (2024). The facilitative effect of the keyword mnemonic on L2 vocabulary retrieval practice. *Heliyon, 10*. https://doi.org/10.1016/j.heliyon.2024.e25212

Soderstrom, N. C. (2026). Learning Through Retrieval Variability: Benefits and Metacognitive Judgments in Foreign-Language Vocabulary Learning. *Behavioral Sciences, 16*. https://doi.org/10.3390/bs16091705

Tuychiev, A. (2026). THE ROLE OF SPACED REPETITION AND RETRIEVAL PRACTICE IN SUPPORTING LONG-TERM L2 VOCABULARY DEVELOPMENT. *SCIENCE TIME*. https://doi.org/10.66345/stj.v4i5/1.6122

Van Den Broek, G. V. D., Takashima, A., Segers, E., & Verhoeven, L. (2018). Contextual Richness and Word Learning: Context Enhances Comprehension but Retrieval Enhances Retention.. *Language Learning, 68*, 546-585. https://doi.org/10.1111/lang.12285

Van Den Broek, G. V. D., Wesseling, E., Huijssen, L., Lettink, M., & Van Gog, T. (2022). Vocabulary Learning During Reading: Benefits of Contextual Inferences Versus Retrieval Opportunities. *Cognitive Science, 46*. https://doi.org/10.1111/cogs.13135